Electric fields

antocciphysics | TEACHER KEY | CPA

Learning target: Use field arrows to predict the force on a test charge.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 18; 3 course days later.*
Bends toward the normal; frequency stays the same at a stationary boundary.

02. Explain a worked example (worked)
Positive charge force follows E; negative charge force opposes E.

03. Look and explain (model)
Field outward; negative test charge force toward the source.

04. Build the solution (practice)
4 N/C right.

05. Use the idea (reason)
No; it probes an existing field, assuming it does not substantially disturb the source arrangement.

06. Choose a method (mixed) *Review: Unit 6, Day 9; 12 course days later.*
60 m; one-way distance is half the round trip.

07. Exit check (exit)
Positive down; negative up.

08. STRETCH: Apply it another way (stretch)
Equal magnitude opposite charges, e.g.±3 C, give±12 N in this supplied field.

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01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 18; 3 course days later.*
Thermal peak becomes 1/3 by Wien's law; sound wavelength becomes 1/5 by λ=v/f. Different models and assumptions.

02. Explain a worked example (worked)
Positive charge force follows E; negative charge force opposes E.

03. Represent and interpret (model)
Field outward; negative test charge force toward the source.

04. Solve and justify (practice)
6.0×10^{-4} N west.

05. Analyze the reasoning (reason)
No; it probes an existing field, assuming it does not substantially disturb the source arrangement.

06. Choose a method (mixed) *Review: Unit 6, Day 9; 12 course days later.*
λ=1.6 m; f≈212.5 Hz.

07. Exit check (exit)
One fourth; isolated point charge or spherically symmetric source outside its extent.

08. STRETCH: Apply it another way (stretch)
Magnitude 4000 N/C downward; divide force by signed charge.